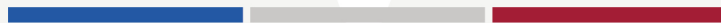




MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 3 UNIT 1
Responsible agentic AI

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Learning outcome:

LO1: Explore the risks, governance, and technical considerations associated with deploying agentic AI systems.

1. Introduction

In this final module of the program, you will explore the questions that determine whether agentic AI can create sustainable value:

- How to scale responsibly?
- How to govern with confidence?
- How to build the organizational infrastructure to sustain AI value creation?

You will also hear from experts on deployment, measurement, and governance as the focus shifts from pilots to enterprise scale.

2. Reframing scaling

Paul McDonagh-Smith opens the module with a clear warning: a strong agentic AI demo is not the same as a scalable system. What looks impressive in a pilot can quickly break down in production, where edge cases, weak oversight, and poor measurement become much harder to manage.

He also argues that scaling is not simply about expanding what already works. It means moving into a more demanding operating environment, where governance, human oversight, and organizational readiness become critical.

Main message: Scale through focused experiments, fast learning, and disciplined judgment, not through ambition alone.



Video 1: The scaling paradox. (Access this set of notes on the Online Campus to engage with this video and download its transcript.)

3. Autonomy spectrum and safeguard architecture

In Video 2, McDonagh-Smith makes the case that autonomy is not a yes-or-no decision. It is something leaders must calibrate task by task, based on context, stakes, and the consequences of error. His key framework is the autonomy spectrum, which runs from tightly constrained systems to highly autonomous ones.

He then identifies four safeguard layers that make this usable in practice:

1. **Hardcoded constraints:** Set non-negotiable limits outside the model.
2. **Output verification:** Check results before they reach a user or trigger an action.
3. **Reversibility by design:** Be able to undo errors before they spread.
4. **Clear escalation paths:** Route decisions to a human when the system reaches its limits.

The broader insight is that agentic systems should not be judged against perfection, but against the current process they are meant to improve.

Main message: Designing for uncertainty is about making risk visible, bounded, and manageable rather than assuming it can be designed away.



Video 2: Designing for uncertainty. (Access this set of notes on the Online Campus to engage with this video and download its transcript.)

4. Connecting design to advantage

In Video 3, McDonagh-Smith shifts from operations to strategy. He argues that agentic AI is changing the economics of enterprise technology by lowering the cost of building and maintaining software. As a result, leaders need to revisit long-held assumptions about build vs. buy, what truly differentiates, and which technology choices will still make sense in an AI-native environment.

He also points to a deeper shift: Software may increasingly be defined by structured intent, business logic, and process design, rather than by code alone. This raises a broader leadership question about which assets will appreciate as AI advances. His closing point is that organizational readiness is now the real constraint.

Main message: The organizations best positioned to scale are the ones that combine top-down sponsorship with bottom-up experimentation and create the conditions for fast, focused learning.



Video 3: From architecture to advantage. (Access this set of notes on the Online Campus to engage with this video and download its transcript.)

Note:

At this point in the lesson, you can test your understanding of the content with a practice quiz. Access this lesson on the Online Campus to begin.

5. Conclusion

Module 3 brings the program to its culminating challenge: turning early agentic AI promise into lasting enterprise value. Across the videos that follow, McDonagh-Smith and his fireside chat guests make several key points clear:

- **Scaling** is not simply expanding a pilot.
- **Autonomy** must be calibrated, not assumed.
- **Long-term advantage** will depend as much on governance, readiness, and organizational learning as on the technology itself.